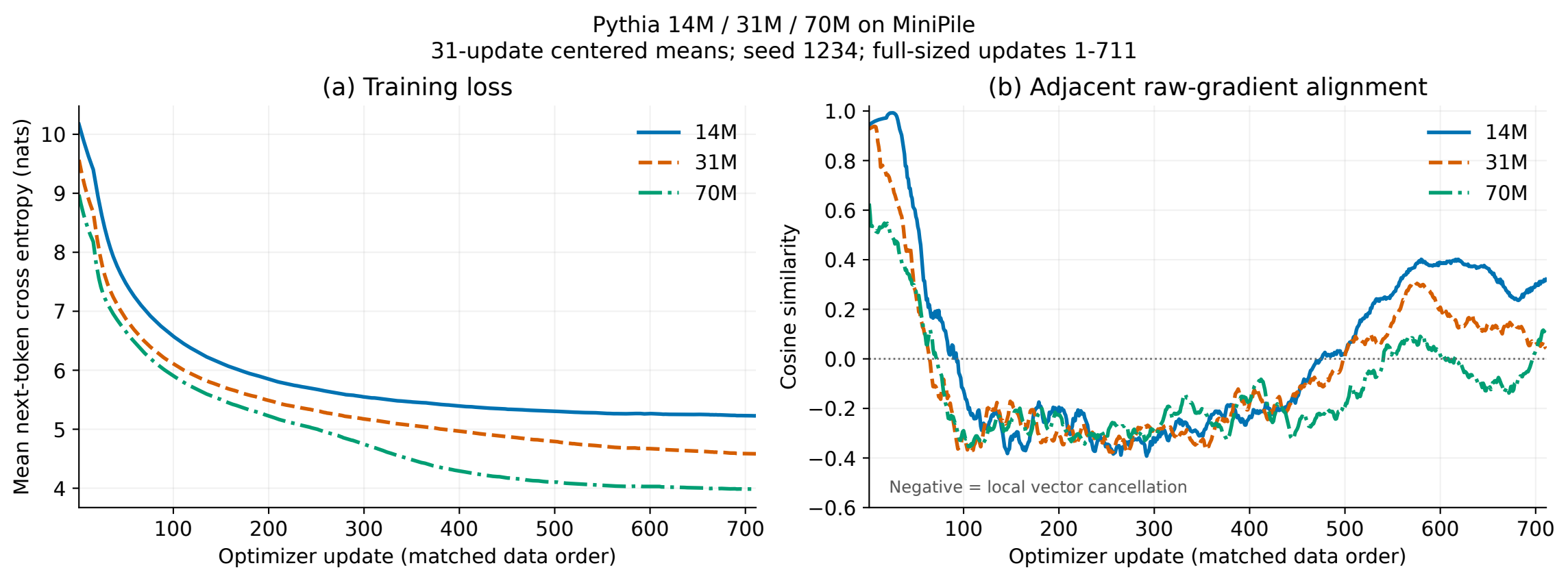

Pythia 14M / 31M / 70M on MiniPile
31-update centered means; seed 1234; full-sized updates 1-711
(a) Training loss
Mean next-token cross entropy (nats)
Optimizer update (matched data order)
14M
31M
70M
(b) Adjacent raw-gradient alignment
Cosine similarity
Negative = local vector cancellation
Optimizer update (matched data order)
14M
31M
70M